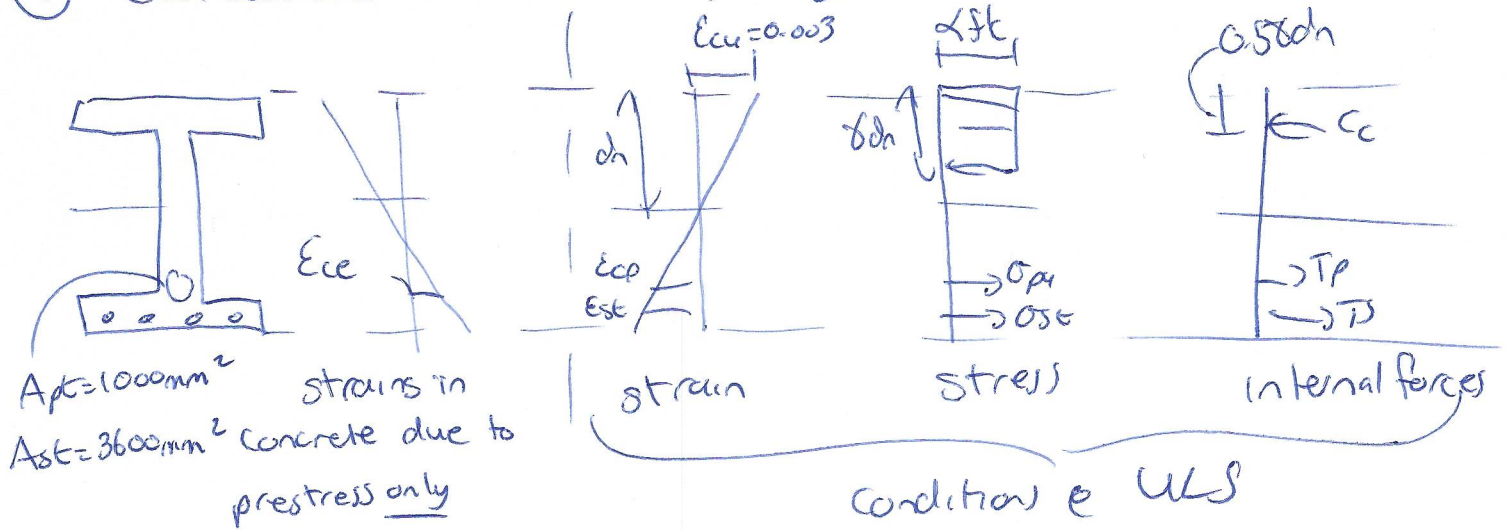


Problem 8 - solutions - Design for flexure

① Calculate moment capacity M_u



$\Rightarrow$ determine d_n

check whether stress block extends into web

Assume steels have yielded

$$T_s = T_{sy} = 3600 \times 500 = 1800 \times 10^3 \text{ N}$$

$$T_p = T_{py} = 1000 \times 1640 = 1640 \times 10^3 \text{ N}$$

Compression force in flange

$$C_f = 0.8 f_{lc} \times t \times b$$

$$= 0.8 \times 32 \times 150 \times 600 = 2304 \times 10^3 \text{ N}$$

αs-1-3
 $f_{lc} = 32$
 $\alpha = 0.8$
 $\gamma = 0.89$

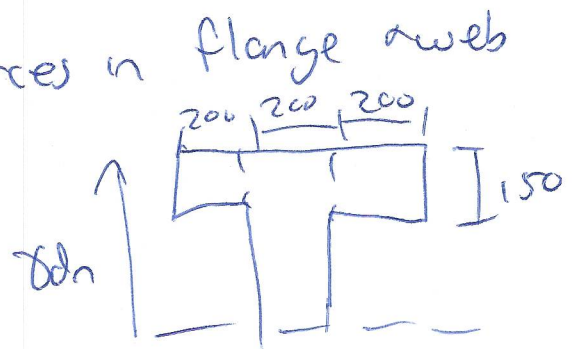
Since $T_p + T_s > C_f \therefore$ stress block must extend into web

* Calculate compressive forces in flange & web

$$C_f = 0.8 f_{lc} \times t (b - b_w)$$

$$= 0.8 \times 32 \times 150 (400)$$

$$= 1536 \times 10^3 \text{ N}$$



$$C_w = 0.8 f_{lc} \times \gamma d_n \times b_w$$

$$= 0.8 \times 32 \times 0.89 d_n \times 200$$

$$= 4557 d_n$$

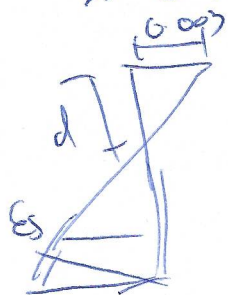
* Force equilibrium

$$C_f + C_w = T_s + T_p$$

$$1563 \times 10^3 + 4557 d_n = 1800 \times 10^3 + 1640 \times 10^3$$

$$d_n = 412 \text{ mm}$$

* Check assumptions steels have yielded.



$$\epsilon_{st} = \frac{\epsilon_{cu} (d_{st} - d_n)}{d_n} = \frac{0.003 (963 - 412)}{412}$$

$$= 0.004 > \epsilon_{sy} (0.0025) \therefore \text{ok.}$$

→ check strain in prestressing steel

$$\text{total strain @ level } d_{PT} = \epsilon_p = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$$

ϵ_{ce} - due to prestress only

$$[P_j = 0.7 \times 184 \times 10 = 1288 \text{ kN}; P_i = 0.85 \times P_j = 1095 \text{ kN}; P_e = 0.9 \times P_i = 986 \text{ kN}]$$

$$\sigma_{ce} = \frac{P_e}{A_g} + \frac{P_e \cdot e^2}{I_g} = \frac{986 \times 10^3}{320,000} + \frac{986 \times 10^3 \times 350^2}{3.86 \times 10^{10}}$$

$$= 6.2 \text{ MPa.}$$

$$\epsilon_{ce} = \frac{\sigma_{ce}}{E_c} = \frac{6.2}{26,100} = 0.00021$$

ϵ_{pe} = strain in prestress steel

$$\epsilon_{pe} = \frac{\sigma_{pe}}{E_p} = \frac{(986,000/1000)}{190,000} = 0.00520$$

ϵ_{cp} - strain in concrete @ level of prestress due to applied M.

$$= \frac{\epsilon_{cu} (d_p - d_n)}{d_n} = \frac{0.003 ((500 + 350) - 412)}{412}$$

$$= \underline{0.0032}$$

Total strain @ level of prestress

$$\epsilon_{pu} = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$$

$$= 0.00021 + 0.0052 + 0.0032$$

$$= 0.00861 \gg \epsilon_{py} = \frac{1640}{200,000} (0.0082)$$

$\therefore \underline{ck}$

$\Rightarrow$ Calculate Moment Capacity, M_u

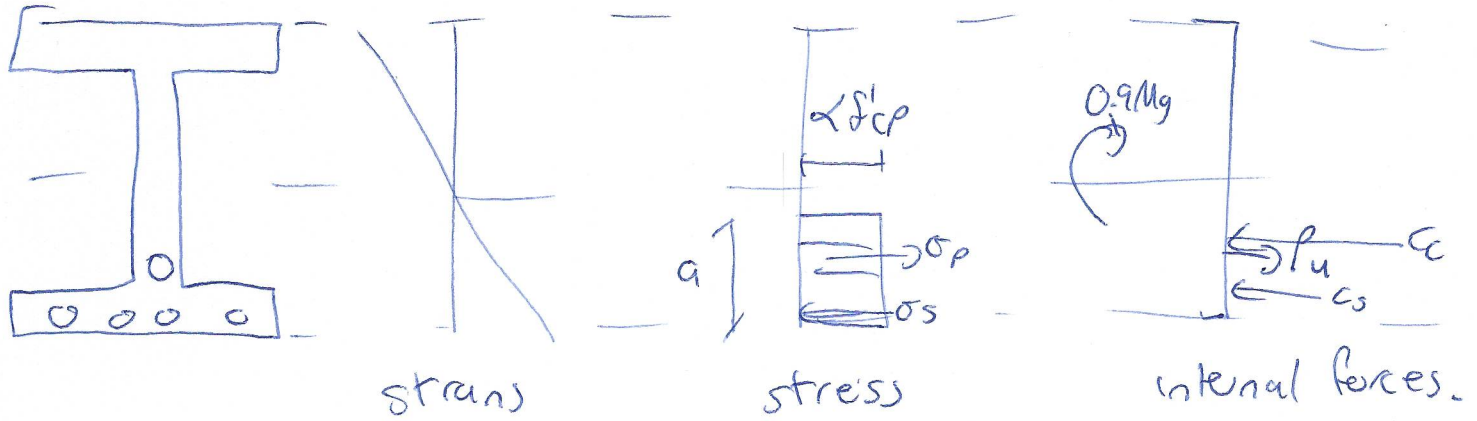
$$M_u = T_s d_s + T_p d_p - C_f d_f - C_{cs} d_{cs}$$

$$= (1800 \times 10^3 \times 963) + (1640 \times 10^3 \times 850)$$

$$- (1536 \times 10^3 \times 75) - \left(4557 \times 413 \times \frac{413 \times 0.89}{2} \right)$$

$$= \cancel{2667 \text{ kNm}} \quad 2667 \text{ kNm} \quad (+10^6)$$

② Calculate P_u & transfer.



$$A_{st} = 3600 \text{ mm}^2, f_{cp} = 25 \text{ MPa}, \alpha = 0.8125$$

$$0.9 \times Mg = 0.9 \times \left(0.32 \times 25 \text{ kN/m}^3 \times \frac{16^2}{8} \right) \quad C_c = C_f + C_w$$

$$= 0.9 \times 256 \text{ kNm}$$

$$C_s = 3600 \times 500 = 1800 \times 10^3 \text{ N}$$

$$C_f = \alpha f_{cp} \cdot d_f \cdot b_f = 0.8125 \times 25 \times 150 \times 600 = 1828 \times 10^3 \text{ N}$$

$$C_w = \alpha f_{cp} \times a \times 200 = 4063 a$$

⇒ Force equilibrium

$$P_u = C_f + C_w + C_s$$

$$P_u = 1828 \times 10^3 + 1800 \times 10^3 + 4063 a$$

$$P_u = 3628 \times 10^3 + 4063 a \quad \text{--- (1)}$$

⇒ Moment equilibrium @ bottom fibre of beam

$$C_f d_f + C_s \times d_s + C_w \times \frac{a}{2} - P_u d_p = 0.9 Mg \text{ (sub in } P_u) \quad \text{--- (2)}$$

$$\left(1800 \times 10^3 \times 37 \right) + \left(1828 \times 10^3 \times 75 \right) + 4063 a \times \frac{a}{2} = \left[\left(3628 \times 10^3 + 4063 a \right) \times 150 \right] + \left(0.9 \times 256 \times 16^6 \right)$$

$$2031.5a^2 - 609450a - 553 \times 10^6 = 0$$

Solving quadratic for a

$$a = 692 \text{ mm} \quad \therefore P_u = 6439.6 \text{ kN}$$

$$0.6 P_u > 1.15 P_j$$

$$0.6 \times 6440 > 1.15 \times 1288$$

$$3864 > 1481 \quad \therefore \text{OK @ transfer}$$

Problem 9 - Solutions - Shear Design

⇒ Design load

$$w^* = 1.2 \times 8 + 1.5 \times 14 \\ = 30.6 \text{ kN/m}$$

$$W_g = 14 \text{ kN/m} \\ S.W_{con} = 25 \text{ kN/m}^3$$

$$V(x) = 245 - 30.6x \text{ (kN)}$$

$$M(x) = 245x - 15.3x^2 \text{ (kNm)}$$

⇒ force in cable and slope @ distance x from support & eccentricity

$$P_g = 0.8 \times 10 \times 184 = 1472 \text{ kN}$$

$$P_c = 0.95 \times 1472 = 1398 \text{ kN} \quad (5\% \text{ immediate losses})$$

$$P_e = 0.8 \times 1398 = \underline{1118 \text{ kN}} \quad (20\% \text{ deferred losses})$$

slope
eccentricity

$$y(x) = 4h \left[\frac{x}{L} - \left(\frac{x}{L} \right)^2 \right] \\ = 4 \times 350 + \left[\frac{300}{16} - \left(\frac{x}{16} \right)^2 \right]$$

$$e = \frac{x}{11.4} - \frac{x^2}{131} \text{ (m)}$$

slope

$$\theta(x) = \frac{4h}{L} \left(1 - 2 \frac{x}{L} \right)$$

$$\theta = 0.0875 \left(1 - \frac{x}{8} \right) \text{ (rad)}$$

① ⇒ shear design at section distance d_v from support

$$d_v = 0.72 \times 1000 = 0.72 \times 1000 = 720 \text{ mm}$$

$$\text{take } x = \underline{0.72 \text{ m}}$$

⇒ Effective width of web

duct diameter = 112 mm

$$b_v = 200 - 0.5 \times 112 = 144 \text{ mm}$$

⇒ vertical component of cable force

$$P_v = P \cdot \alpha = 1118 \times 0.0796 \\ = 89 \text{ kN}$$

$$V^* = 245 - 20.6 \times 0.72 = 223 \text{ kN}$$

$$M^* = 245 \times 0.72 + 15.3 (0.72)^2 = 168 \text{ kNm}$$

⇒ Calculate concrete contribution to shear capacity

cl 8.2.4.1

$$V_{uc} = k_b b_v d_v \sqrt{f_c} \quad K_v = \left[\frac{0.4}{1 + 1500 \epsilon_x} \right]$$

ϵ_x = longitudinal strain in concrete for shear - cl 8.4.2.2.

$$\epsilon_x = \frac{\left[\frac{M^*}{d_v} + V^* \right] + \gamma_p P_v + 0.5 V^* - A_{pt} f_{po}}{2 [E_s A_{st} + \gamma_p A_{pt}]} \quad \gamma_p = 0.9$$

within $0 \leq \epsilon_{sc} \leq 3 \times 10^{-3}$

$$f_{po} = 0.5 f_{pb}$$

$$\epsilon_x = \frac{\left[\frac{168 \times 10^3}{720} + 223 \times 10^3 \right] + 0.9 \times 89 \times 10^3 - 1000 \times 0.5 \times 1870}{2 [200,000 \times 3600 + 190,000 \times 1000]}$$

$$= 0.0003$$

since calculate value of E_x is less than zero, we shall recalculate using

$$E_x = \frac{\left[\frac{M^*}{d_v} + V^* \right] + \gamma_p P_v + 0.5 N^* - A_{pt} f_{pb}}{2 \left[E_s A_{st} + E_p A_{pt} + E_c A_{ct} \right]}$$

$$A_{ct} = \frac{1000}{2} \times 200 + 150 \times (600 - 200) = 145,000 \text{ mm}^2$$



$$E_c =$$

$$E_x = \frac{-358767}{2 \left(9.1 \times 10^8 + (145 \times 10^3 \times 30,100) \right)} = -32.97 \times 10^{-6} > -0.2 \times 10^{-3}$$

$$\therefore K_v = \frac{0.4}{1 + 1500(-32.97 \times 10^{-6})} = 0.43$$

$$\begin{aligned} \theta_v &= (29 + 7000 E_x) - \text{cl } 8.2.4.2(1) - \text{equation} \\ &= 29 + 7000(-32.97 \times 10^{-6}) \\ &= \underline{28.7^\circ} \end{aligned}$$

$$\begin{aligned} \Rightarrow V_{uc} &= K_v b_v d_v \sqrt{f_k} \\ &= 0.43 \times 144 \times 720 \times \sqrt{32} \quad (\times 10^{-3}) \\ &= \underline{252 \text{ kN}} \end{aligned}$$

$$\phi V_{uc} = 0.75 \times 252 = \underline{189 \text{ kN}}$$

⇒ Design for shear reinforcement

The minimum shear that is required to take by stirrups

$$V_{us} \geq \frac{1}{\phi} [V^* - (\phi V_{kc} + \phi p l_v)]$$

$$\geq \frac{1}{0.75} [223 - (189 + 0.9 \times 89)]$$

$$V_{us} \geq -61.3$$

Since $V_{us} < 0$ ∴ only min shear req needed
as $D > 750 \text{ mm}$

- minimum shear req.

$$\frac{A_{s,min}}{s} = 0.08 \sqrt{f_{lc}} \times \frac{b_v}{f_{sy.f}}$$

take R10
as ligatures

$$\frac{A_{s,min}}{s} = 0.08 \sqrt{32} \times \frac{144}{250}$$

$$A_{sr} = 2 \times 80 = 160 \text{ mm}^2$$

$$\frac{A_{s,min}}{s} = 0.126$$

$$s = 615 \text{ mm}, \text{ however max. spacing} = 300 \text{ mm}$$

∴ adopt R10-300 centres.

- check web crushing - Cl 8.2.3.3

$$V_{u,max} = 0.55 \left[0.9 f_{lc} b_v d_v \left(\frac{\cot \alpha_v + \cot \alpha_v}{1 + \cot^2 \alpha_v} \right) \right]$$

$$\alpha_v = 28.7^\circ$$

$$\alpha_v = 28.7^\circ$$

(angle between longitudinal
req & transverse req)



$$= 0.55 \left[0.9 \times 32 \times 144 \times 720 \left(\frac{1.83 + 0}{1 + (1.83)^2} \right) \right]$$

$$= 899 \text{ kN} > (V^* - \phi p l_v)$$

web crushing not critical

∴ adopt R10-300 cts

② shear design @ $x = 4 \text{ metres}$

@ $x = 4 \text{ m}$, $P_e = 1118 \text{ kN}$ $\alpha = 0.044$

$V^* = 245 - (30.6 \times 4) = 123 \text{ kN}$ $e = 229 \text{ mm}$

$M^* = 245 \times 4 - 15.3 \times (4)^2 = 735 \text{ kNm}$

Vertical component of Prestress

$P_{Vz} = P_e \times \alpha = 1118 \times 0.044 = 49 \text{ kN}$

$b_{Vz} = 144 \text{ mm}$

effective depth of section $d = \frac{A_{st} \cdot d_{st} + A_p \cdot d_p}{A_{st} + A_p}$ @ $x = 4 \text{ m}$

$d = \frac{3600 \times 950 + 1000 \times (500 + 229)}{3600 + 1000} = 901 \text{ mm}$

internal lever arm

$d_v = \text{maximum } (0.72d, 0.9d)$
 $= (0.72 \times 1000) \text{ or } (0.9 \times 901)$

$d_v \text{ take} = 811 \text{ mm}$

⇒ Calculate concrete contribution to shear capacity
cl 8.2.4.1 $V_{uc} = K_v b_v d_v \sqrt{f_c}$

$K_v = \frac{0.4}{[1 + 1500 \epsilon_{xc}]}$

$$\epsilon_x = \frac{\left[\frac{M^*}{I_v} + V^* \right] + \phi_p P_v + 0.5 N^* - A_{pt} f_{po}}{2 [E_s A_{st} + E_p A_{pt}]}$$

$$\epsilon_x = \frac{\left[\frac{735 \times 10^6}{811} + 123 \times 10^3 \right] + (0.9 \times 49) - (1000 \times 0.5 \times 1870)}{2 [200,000 \times 3600 + 190,000 \times 1000]}$$

$$= \frac{94332}{1.82 \times 10^9} = 5.18 \times 10^{-5} \quad \text{within} \quad 0 \leq \epsilon_x < 3 \times 10^{-3}$$

$$K_v = \frac{0.4}{1 + 1500 \times 5.18 \times 10^{-5}} = 0.37$$

$$\theta_v = 29 + 7000 \times 5.18 \times 10^{-5} = 29.4^\circ$$

$$\therefore V_{uc} = 0.37 \times 144 \times 811 \times \sqrt{32} \quad (\times 10^{-3})$$

$$= 244 \text{ kN}$$

$$\phi V_{uc} = 183 \text{ kN} \quad (\phi = 0.75)$$

$V^* \leq \phi V_{uc}$ $\therefore$ min shear req only need. $D > 750 \text{ mm}$
from above adopt R10-300 cts